

Name _____ Date _____ Period _____

Solving Absolute Value Linear Inequalities - Practice

Algebra II Honors | Unit 2 | Lesson 7 Practice | TEK A2.6.F

Objective: I can solve absolute value linear inequalities requiring a sign flip or a variable on both sides, and justify results using distance.

Section 1 — Recall

1. Solve $|x| < 10$, and explain using distance why the answer is $-10 < x < 10$.

_____ ; _____

2. Solve $|x| \geq 6$, and explain using distance why the answer is two separate pieces.

_____ ; _____

Section 2 — Apply

3. Solve $-2|x - 3| + 8 > 0$.

4. Solve $-3|x + 2| + 5 \leq -10$.

Section 3 — Challenge

5. Solve $4 - 2|3x - 1| \geq -6$.

6. Solve $|2x - 5| + 3 < 2x$.

Section 4 — Explain It

7. Using the fact that $|x| \geq 0$ always, explain why $|x| < k$ only has solutions when $k > 0$, and describe what happens when $k \leq 0$.
